

Jayben Bushnell

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[LinkedIn](#) | [Github Portfolio](#) | Native English - Proficient Japanese

Staff Unity Engineer

10+ Years Professional Unity 3D Game Engineering • Live-Service Mobile at 20M+ DAU Scale • Technical Architecture Leadership & Team Scaling • Prototype-to-Production Ownership • Blockchain & Crypto Gaming Enthusiast

Professional Summary

Staff-level Unity 3D engineer with 10+ years of hands-on experience leading live-service mobile games from early prototype and rapid iteration through full production, launch, and ongoing live operations at massive scale (20M+ downloads / millions of DAU). Excel at providing technical direction, making high-impact architectural decisions, enforcing best practices, conducting code and architecture reviews, and scaling small teams while collaborating cross-functionally with design, art, production, and executive stakeholders. Deep expertise in performant 3D systems, custom physics, real-time multiplayer, F2P monetization, and cloud-native backend services. Passionate about blockchain technology, crypto economies, and the intersection of Web3 with mobile gaming; self-studied blockchain systems, smart contracts, NFTs, and play-to-earn mechanics and actively follow industry trends. Proven “roll-up-your-sleeves” mindset with a track record of shipping polished titles under tight deadlines and earning rare executive praise for code quality and maintainability.

Work History

Senior Software Engineer / HypGames, Inc. - Remote /
Apr 2021 - Present (5 years)

Lead Unity full-stack engineer and technical owner for high-revenue live-service F2P mobile 3D golf titles serving millions of daily active users.

Provided technical direction, architectural decisions, code reviews, and team leadership while scaling features from prototype concepts to production at 20M+ scale.

- **Ultimate Golf! (20M+ Downloads)** [Android](#), [iOS](#)

- Led design and implementation of new 3D gameplay systems and features from initial prototype to production, including building an entirely new custom physics engine for the Lucky Putt / Mini-Golf mode under extreme time pressure while collaborating with design, art, and QA.
- Architected highly scalable modular frameworks using Factory and Static Service patterns that enabled rapid iteration and long-term maintainability for monetization, social, and gameplay systems at massive concurrency.
- Designed and launched company-wide reusable social systems (Guilds) supporting real-time synchronization for thousands of concurrent players; later adopted across nearly all Hyp titles.
- Built watch-to-earn F2P monetization systems with versioned APIs and persistent data handling that generate thousands in daily revenue from non-paying users with zero-downtime backward compatibility.
- Owned performance optimization and low-level engineering: implemented C++ native DLLs, deep multi-threaded concurrency tuning, and regular memory profiling that delivered major reductions and buttery-smooth gameplay on low-end devices.
- Created large-scale automated testing infrastructure with thousands of bots to simulate real-world production stress and catch issues early.
- Provided technical leadership and mentorship to a team of 3 engineers on major features (spectating, guilds, mini-golf); drove on-time delivery through empathy-driven conflict resolution and architecture reviews.
- Maintained Java backend services on AWS ECS/EC2 with TypeScript admin tools; handled localization, live events, and zero-downtime migrations supporting 20M+ users.

- **PGA Tour Pro Golf (4.8/5) [Apple Arcade](#)**

- Owned and refactored the complete 3D client codebase into a production-grade, cross-platform Unity framework; delivered months ahead of Apple's strict deadlines (one of the fastest approvals they had seen) with rare executive praise for code quality, maintainability, and polish.
- Engineered responsive landscape/portrait 3D UI systems with safe areas and adaptive scaling, enabling single-codebase deployment across phones, tablets, PCs, and TVs.
- Developed Objective-C native bridge for complex runtime emoji rendering and seamlessly integrated it into Unity multiplayer chat and player systems.
- Led multiple engineers and conducted architecture reviews while mentoring juniors to improve code quality and delivery velocity.

Unity Game Developer / Contec Games. - Remote / March 2016 - Apr 2021 (5 years 1 month)

Shipped King of the World – full client-server 3D multiplayer real-time strategy title on Steam. Owned end-to-end development in Unity/C# from prototype to launch.

- **[King of the World - Steam](#)**

- Engineered real-time multiplayer architecture (client-side prediction + lockstep protocol over UDP/RPC) that delivered smooth gameplay for tens of thousands of 3D entities under high latency.
- Built reusable in-game map editor for user-generated content and integrated Steamworks.net for monetization, cosmetics, and achievements.
- Deployed and managed production multiplayer servers on AWS EC2 Linux instances, gaining deep hands-on experience in client-server architecture and reliable live operations.

Solo Game Developer / Self-Taught Developer / 2009 – 2016 (7 years)

- Built and self-published multiple mobile titles to Apple App Store and Google Play; developed core Unity 3D/C# multiplayer skills and self-hosted production servers on AWS.

Skills

- **Core Technologies:**
 - Unity Engine (3D), C# (.NET), Java (backend/server), Objective-C (native bridging), C++
- **Architecture & Leadership:**
 - System architecture & design patterns (Factory, Static Service), technical direction, code/architecture reviews, team scaling & mentoring, best-practice enforcement
- **Gameplay & Production:**
 - 3D game systems, custom physics engines, rapid prototyping to full production, live-service tooling, cross-platform UI/UX, F2P monetization
- **Networking & Backend:**
 - Real-time multiplayer, client-side prediction, lockstep protocol, UDP/RPC, AWS ECS & EC2, persistent data versioning, zero-downtime migrations
- **Performance:**
 - Memory profiling & major reductions, deep concurrency tuning, mobile optimization at scale
- **Other:**
 - Git, Steamworks.net, Datadog, TypeScript admin tools; Passion for blockchain/crypto systems, NFTs, Web3 gaming economies, and play-to-earn mechanics
- Additional Information:

- Open to relocation for the right opportunity (please see cover letter).
- Portfolio includes live 3D gameplay systems, custom physics demos, guild/social frameworks, and large-scale multiplayer architecture examples. See my [github portfolio page](#)